| No. | Status | Date |
|---|---|---|
| LNDE-RPT-20241202-036-DS-6-RA-103 | Issued | 2024-12-10 |

# UT Thickness Data Sheet - 6-RA-103

## Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for the rich amine header 6-RA-103 located within the ATU-100. The scope of work is limited to the verification of wall thickness at designated condition monitoring locations to assess asset integrity and remaining life.

Marcus Chen, acting as the lead UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and data recording. All activities were conducted in accordance with the requirements specified in QSCP-NDT-001 Rev 3.

## Technical Basis and Standards

Thickness measurements were executed using ultrasonic testing techniques as defined in ASTM E797. The methodology for data acquisition, including the CML-id format and digital export requirements, follows the site-specific guidelines established in QSCP-NDT-001 Rev 3.

The results presented in this data sheet supplement the findings detailed in LNDE-RPT-20241202-036-R0, which provides the comprehensive PAUT corrosion mapping for the associated piping components.

## Line Specifications

The subject line, 6-RA-103, is constructed from A106 Gr.B with a nominal outer diameter of 168.3 mm and a nominal wall thickness of 10.97 mm. The design parameters for this service are 862 kPa(g) and 82 °C.

The minimum allowable thickness (t-min) for this line is 4.57 mm. Any reading recorded below this threshold is subject to the site's non-conformance reporting process.

## Thickness Measurements and Analysis

At CML 6-RA-103-11, the measured wall thickness is 5.82 mm, resulting in a calculated remaining life of 2.1 years. This data is consistent with the mapping performed in LNDE-RPT-20241202-036-R0.

At CML 4-RA-101-06, the measured thickness is 3.55 mm. As this value is below the t-min of 4.57 mm, the remaining life is recorded as expired.

## Thickness readings, measured 2024-12-10

**Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-101-06 | 8.56 | 3.81 | 2024-11-04 | 3.59 | 3.55 | 0.426 | 0.659 | expired | immediate | < t-min; gauge photo attached |

**Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-RA-103-11 | 10.97 | 4.57 | 2024-11-04 | 5.92 | 5.82 | 0.419 | 0.600 | 2.1 | 2025-12-25 | RL/2 < interval; gauge photo attached |

| Sign-off | Name | Function | Date | Signature |
|---|---|---|---|---|
| Prepared | Marcus Chen | lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II | 2024-12-10 | _____ |